

Machine Learning-Guided Modulation of Li⁺ Solvation Structures towards Optimal Electrolyte Systems for High-Performance Li–O₂ Battery

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Abstract: As the “blood” of Li–O₂ batteries (LOBs), electrolytes with various solvation structures of Li⁺ can greatly influence the composition of solid electrolyte interphase (SEI) on anode and the growth kinetics of Li₂O₂ on cathode, and further the battery performance. However, achieving delicate modulation of the multi-composition electrolytes remains significantly challenging to simultaneously give consideration to both the anode and cathode reactions. In this work, we employed Bayesian optimization to develop advanced electrolytes for LOBs, enabling the formation of a stable inorganic-rich SEI, and modulation of Li₂O₂ morphologies. Thus obtained LOBs using the optimized dual-solvent electrolyte could deliver a discharge capacity of 14,063 mAh g^{−1} at a current density of 500 mA g^{−1}, which is far higher than those using the single-solvent electrolytes. This study not only highlights the critical role of the solvation structure for improving the battery performance, but also provides new insights and important theoretical guidance for delicate modulation of electrolyte compositions.

The development of high-energy-density electrochemical energy storage systems has been rapidly driven by the ever-growing demands, aiming at prolonging the operational lifetime of portable electronic devices and extending the driving mileage of electric vehicles. Among various candidates, Li–O₂ batteries (LOBs) have garnered substantial attention in recent years due to their high theoretical specific energy (~3,500 Wh kg^{−1}).^[1] However, LOBs still face significant challenges such as low practical capacity and high overpotential during charge–discharge cycles, arising from Li dendrites on anode and sluggish reaction kinetics on cathode.^[2] Essentially, there should exist a delicate balance

between the formation of the solid electrolyte interphase (SEI) and the growth of Li₂O₂, which is closely related to the Li⁺ solvation/desolvation process. That is the reason why electrolyte optimization has played a decisive role in boosting battery performance besides anode modification and cathode catalyst design.

Specific solvation structures of Li⁺ can markedly improve the stability of SEIs during cycling, in which the one of “anion-induced ion-solvent-coordinated” (AI-ISC) structure shows particular promise.^[3] In AI-ISC, anions could remain a higher concentration around Li⁺ rather than solvent molecules to form a solvation mode dominated by contact ion pairs (CIPs) and aggregates (AGGs). This configuration will be apt to the derivation of an inorganic-rich SEI, enabling high mechanical strength and excellent chemical stability. Such an inorganic-rich SEI can thus effectively suppress Li dendrite growth, minimize side reactions, and reduce interfacial impedance to enhance Li⁺ transport efficiency.^[4] As for cathode, it will experience first the reduction of O₂ into O₂[−] during discharge, and the following formation of LiO₂ (Li⁺ + O₂[−] → LiO₂) and then its disproportionation (LiO₂ → Li₂O₂ + O₂). The solvation structure of Li⁺ could show significant influence on the formation of LiO₂, while LiO₂ determines the growth kinetics of Li₂O₂ and further the discharge capacity and cycling stability. Undoubtedly, the high-performance LOBs should be under the premise of precise control on the solvation structure of Li⁺.^[5]

To achieve the AI-ISC solvation structure, complex solvent systems with varying donor numbers (DN) have been employed to create a locally high-concentration electrolyte (LHCE). It is proposed that the combination of high- and low-DN solvents could form LHCE structure to greatly improve battery performance. The high-DN solvents like dimethyl sulfoxide (DMSO) could show strong solvation ability, enhancing the solubility of Li⁺ and ion conductivity, while stabilizing SEIs and suppressing dendrite growth. However, the excessive usage of high-DN solvents would result in severe corrosion of Li metal.^[6] As for the low-DN solvents such as tetraethylene glycol dimethyl ether (TEGDME), it is generally used to optimize the electrolyte's fluidity, increasing the ion diffusion rate and reduced solvation ability.^[7] Besides, low-DN solvents with weak coordinating ability can avoid excessive dissolution of Li⁺ and facilitate its desolvation. As reported, AI-ISC can reduce the generation/disproportionation rate of LiO₂, in favor of Li₂O₂ growth into larger sizes followed by the solution growth pathway. Well accepted that the morphology and size

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of Li_2O_2 are crucial in determining the overall discharge capacity, because larger Li_2O_2 particles are generally associated with improved conductivity and higher capacity retention during discharge/charge. The key, of course, is how to realize fine modulation of electrolyte compositions, which regulates both the Li^+ solvation/desolvation process, LiO_2 disproportionation, and the growth pathways of Li_2O_2 .^[8]

In this study, the dual-solvent (DMSO/TEGDME) and single-salt (lithium bis((trifluoromethyl)sulfonyl)azanide, LiTFSI) electrolyte system has been successfully developed by aid of Bayesian optimization (BO), which is a kind of classical machine learning algorithms. Through this data-driven approach, we rapidly pinpoint the Pareto front of performance indexes to identify the optimal electrolyte formulation that could establish a careful balance between capacity and overpotential during cycling. The optimized electrolyte enables the LOBs with exceptional discharge capacity and long cycle life, significantly improving the overall battery performance. On anode, the interaction between anions and solvent molecules suppresses solvent reduction, thereby facilitating the formation of inorganic-rich SEIs, and meanwhile significantly improving the Li deposition/stripping behavior. On cathode, it leads to the formation of bigger sized Li_2O_2 with a well-defined hierarchical morphology that favors enhancing the discharge capacity under high current densities. In combination with density functional theory (DFT) and molecular dynamics (MD) simulations, it has provided us a completely new understanding on Li^+ solvation structures towards battery performance from a collaborative work perspective of electrolyte, anode, and cathode.

Figure 1a presents the roadmap of BO to optimize the electrolyte formulation of DMSO/TEGDME/LiTFSI. By adjusting the ratio of DMSO/TEGDME (x_1) and the concentration of LiTFSI (x_2), we modified the composition in order to pursue a higher discharge capacity as $f(x_1)$ but at a lower charge–discharge overpotential as $f(x_2)$. From the functional plots (Figures 1b and c), it can be observed that there is no clear correlation between performance indexes and electrolyte formulation, highlighting the difficulty of optimization. However by constructing the Pareto front during iterations,^[9] the optimal composition could be finally determined to be DMSO/TEGDME of 97.6 %/2.4 % and LiTFSI of 0.47 mol/L, which was denoted as DT (Figure 1d and S1). It is worth mentioning that the optimized DT sample has delivered the longest lifetime among the currently reported carbon-based LOBs (Figures 1e and S2).^[2b,5b,8a,10]

MD simulations (Figures 2a and S3–5) show that TFSI[−] anions will enter the Li^+ solvation shell, forming predominant AI-ISC structures composed of CIPs and AGGs in all the three electrolytes of TEGDME, DMSO, and DT. However, an emerging unique structure of $2\text{Li}^+-4\text{DMSO}-3\text{TFSI}^-$ (Figure 2b) indicates the relatively stronger interaction between Li^+ and the solvent components in DT, and meanwhile TEGDME is located at the second solvation shell judged by DFT (Figure 2c). The differential charge analysis firmly reveals the significant electron density transfer from TEGDME to the first solvation shell of TFSI[−]. The ^7Li nuclear magnetic resonance (NMR) spectra (Figure 2d) provides us more experimental evidences on the solvation structure. By compar-

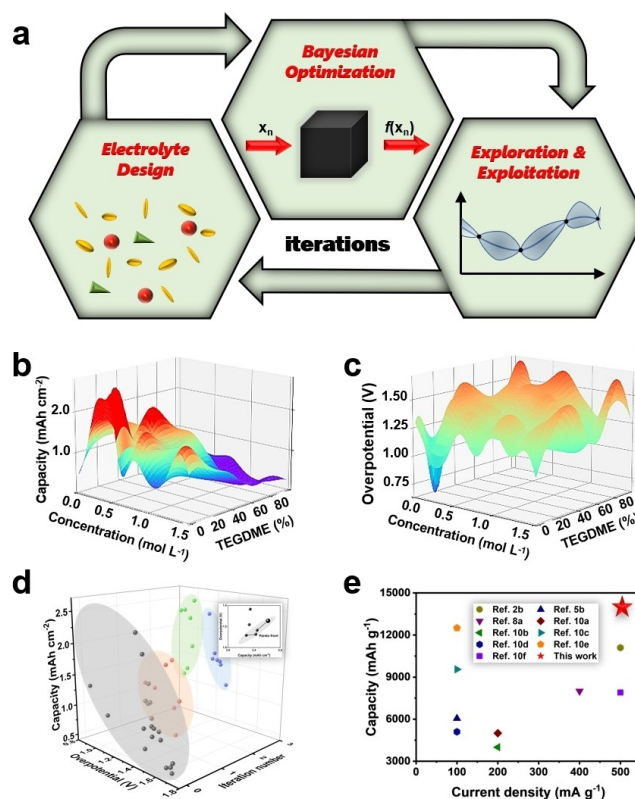


Figure 1. (a) Roadmap of machine learning; (b) Discharge capacity as a function of electrolyte composition; (c) Overpotential as a function of electrolyte compositions; (d) Pareto front of high discharge capacity and low overpotential (black, red, green, and blue represent the 1st, 2nd, 3rd, and 4th rounds of iterations, respectively, and the asterisk marks the selected optimal ratio); (e) Comparison on battery performances of carbon-based LOBs.

ing the curves of the DT and DMSO based electrolytes, the upfield shift in DT should be attributed to the enhanced coordination of Li^+ with DMSO molecules, suggesting higher electron density around Li^+ .^[11] Raman and IR spectra (Figures S6 and S7) can additionally support the solvation structure that the characteristic S–N–S stretching vibration of TFSI[−] clearly shifted to higher wavenumbers, firmly indicating the presence of more AGGs in DT when compared to the DMSO and TEGDME based electrolytes.^[12] The activation energies of Li^+ desolvation are measured to be 7.99, 12.75, 4.50 kJ mol^{−1} for TEGDME, DMSO, and DT, respectively (Figure 2e and S8). The lower energy of DT means its faster Li^+ desolvation kinetics (Figure S9–11). The linear sweep voltammetry (LSV) results (Figure 2f) show that DMSO decomposed at an oxidation potential of 3.3 V, however, it could remain stable up to 4.7 V in DT. Such a significant increase underscores the superior oxidative stability of the DT electrolyte. In addition, we measured the DN of the DT electrolyte solvent to be 29.6 kcal mol^{−1} using ^{23}Na NMR, which is very close to that of DMSO (Figure S12). We therefore anticipate that DT electrolyte will induce a solution-mediated growth mechanism.

In the following, we will discuss the interaction relationship between anode and electrolyte. Note that, Li metal should be

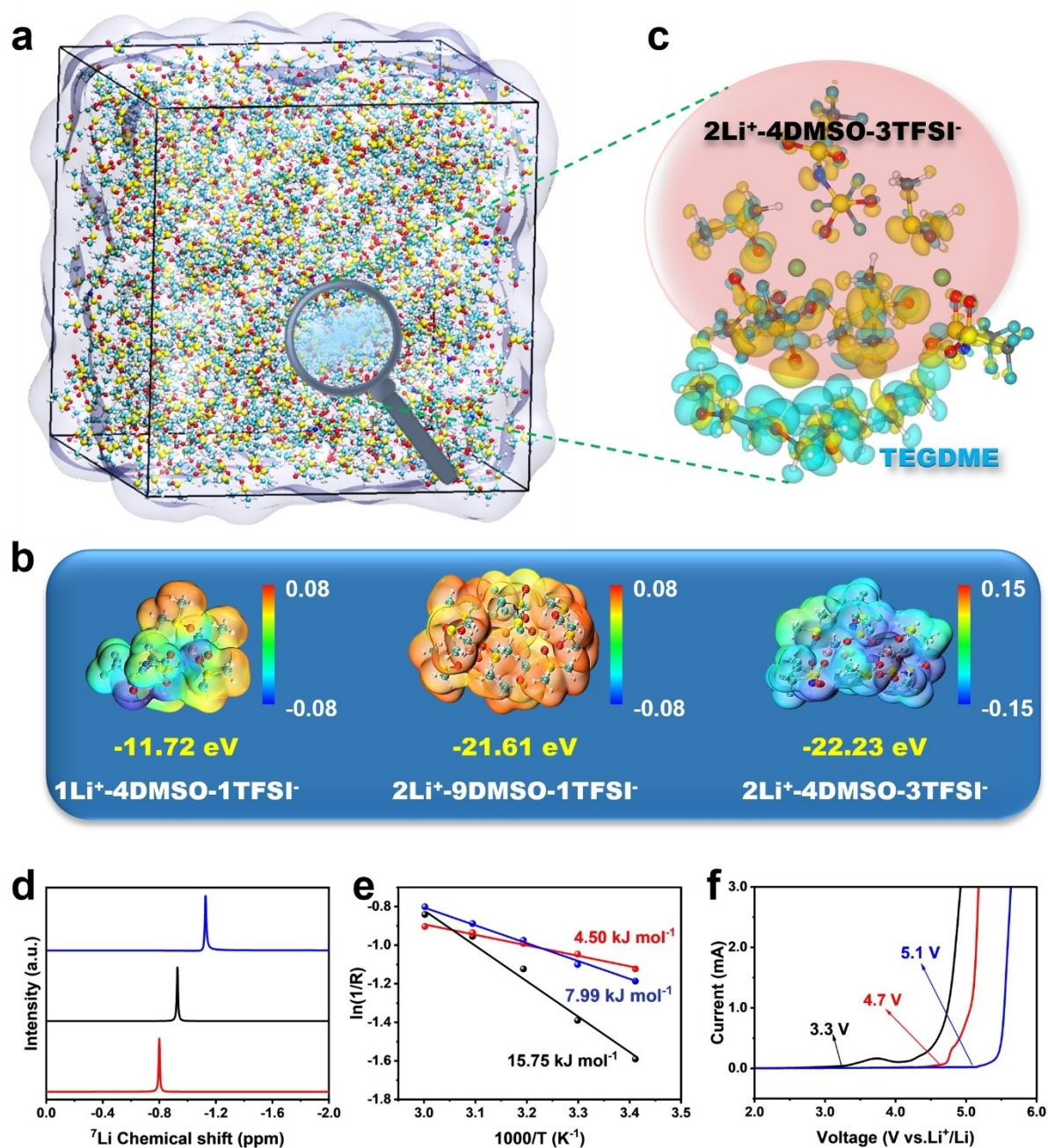


Figure 2. (a) Snapshot of the MD simulation box, (b) possible solvation structures and electrostatic potentials, and (c) differential charge maps in the DT electrolyte (blue for electron loss, yellow for electron gain); (d) ^7Li NMR spectra, (e) activation energy of Li^+ desolvation, and (f) LSV curves (The blue, black, and red lines represent data from TEGDME, DMSO, and DT, respectively.)

oxidized into Li^+ on anode, so Li^+ must pass through SEIs into electrolyte. SEM images reveal that the DMSO sample shows a gully morphology, and even more serious dendrite formation could be observed in TEGDME (Figure 3a, 3b, S13, and S14). However, in sharp contrast, DT has a much thinner, denser, and smoother surface (Figure 3c and S15). X-ray photoelectron spectroscopy (XPS) was then conducted to achieve more detailed compositional information of SEIs (Figures 3d, S16, and S17). The SEI formed in the DT electrolyte initially exhibited a higher C content compared to

those in DMSO and TEGDME before etching, featured by an organic-rich layer. As the etching depth increased, the C content significantly decreased and the content of inorganic components in all three electrolytes increased significantly, and moreover the O 1s spectra revealed that the characteristic of the Li_2O -rich SEIs.

The other inorganic components, such as LiF and Li_2S , progressively increased as well, exhibiting the mosaic characteristic of the SEIs. These findings disclose that the SEI formed in the DT electrolyte has an outer organic-rich layer that

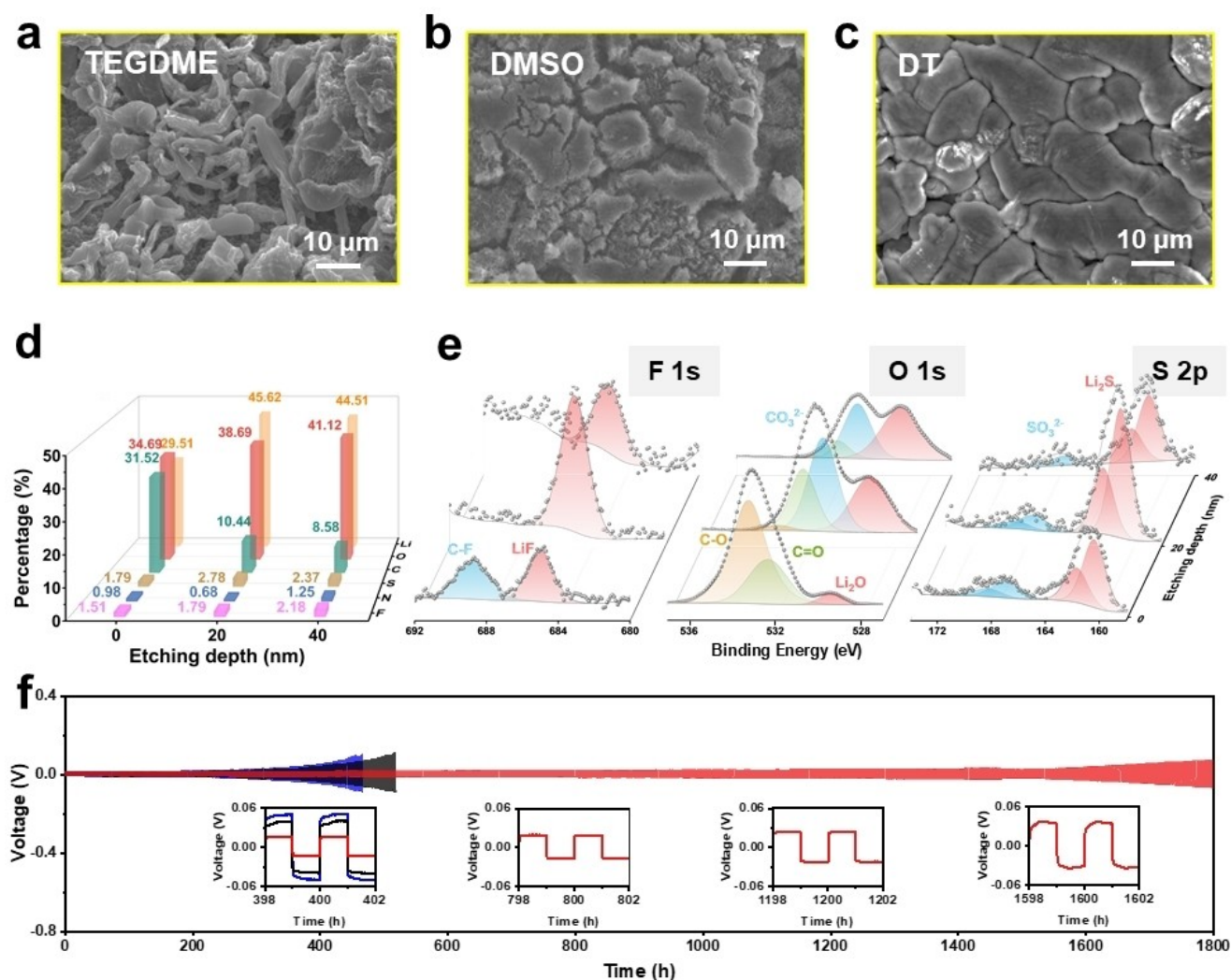


Figure 3. The morphologies of Li anodes obtained by using (a) TEGDME, (b) DMSO and (c) DT after depositing at 0.1 mA cm^{-2} for 5 mAh cm^{-2} ; (d) F 1s, O 1s, S 2p XPS spectra of SEI formed by using DT; (e) Content of different elements in DT at different depths of Ar^+ etching; (f) Cycling performance of Li | Li symmetric cells in different electrolytes (blue, black, and red lines represent data from TEGDME, DMSO, and DT, respectively).

enhances flexibility and an inner inorganic-rich layer that promotes Li^+ conduction and suppresses Li dendrite growth.^[13] The Li | Li symmetric cells are further tested to monitor the long-term cycling states of Li plating and stripping, as shown in Figure 3f. The DT-based electrolyte cells can deliver stable cycles for over 1800 h, and there is little lithium dendrite growth, as shown in Figure S18.

As for cathode, we have investigated the morphologies of discharge products in LOBs, and tried to discuss the possible reaction mechanisms. As reported, the traditional electrolytes, including TEGDME and DMSO, could result in the generation of Li_2O_2 primarily through the solution and the surface growth pathways.^[14] As seen in Figures 4a and b, Li_2O_2 particles are in a small size in the TEGDME and DMSO electrolytes, which have been compactly aggregated on cathode. In contrast, DT results in a much larger size of Li_2O_2 particles that follows by a solvation-mediated mechanism (Figure 4c and S19).^[15] The piles of larger particles greatly

reduce the coverage of active sites on cathode, improving electron transport and oxygen diffusion, thereby enhancing discharge capacity. However, after the full discharge of LOBs, electrochemical impedance spectroscopy (EIS) results show that despite the larger size of Li_2O_2 particles, the impedance of the DT-based LOBs is lower than those of the TEGDME- and DMSO-based batteries (Figure S20). This contributes to a reduced charge overpotential. In situ differential electrochemical mass spectrometry (DEMS) (Figures 4d–f) reveals that the charging process of LOBs by using DT approaches a two-electron reaction, demonstrating fewer side reactions, allowing LOBs to maintain stability over prolonged cycling.^[16] As seen in the radar map (Figure 4g), DT shows excellent ion conductivity, electrochemical stability, and desolvation capability, resulting in greatly enhanced discharge capacity and reduced overpotential. Thanks to these advantages, the DT-based LOBs exhibit excellent rate capability and electrochemical performance when compared to TEGDME and

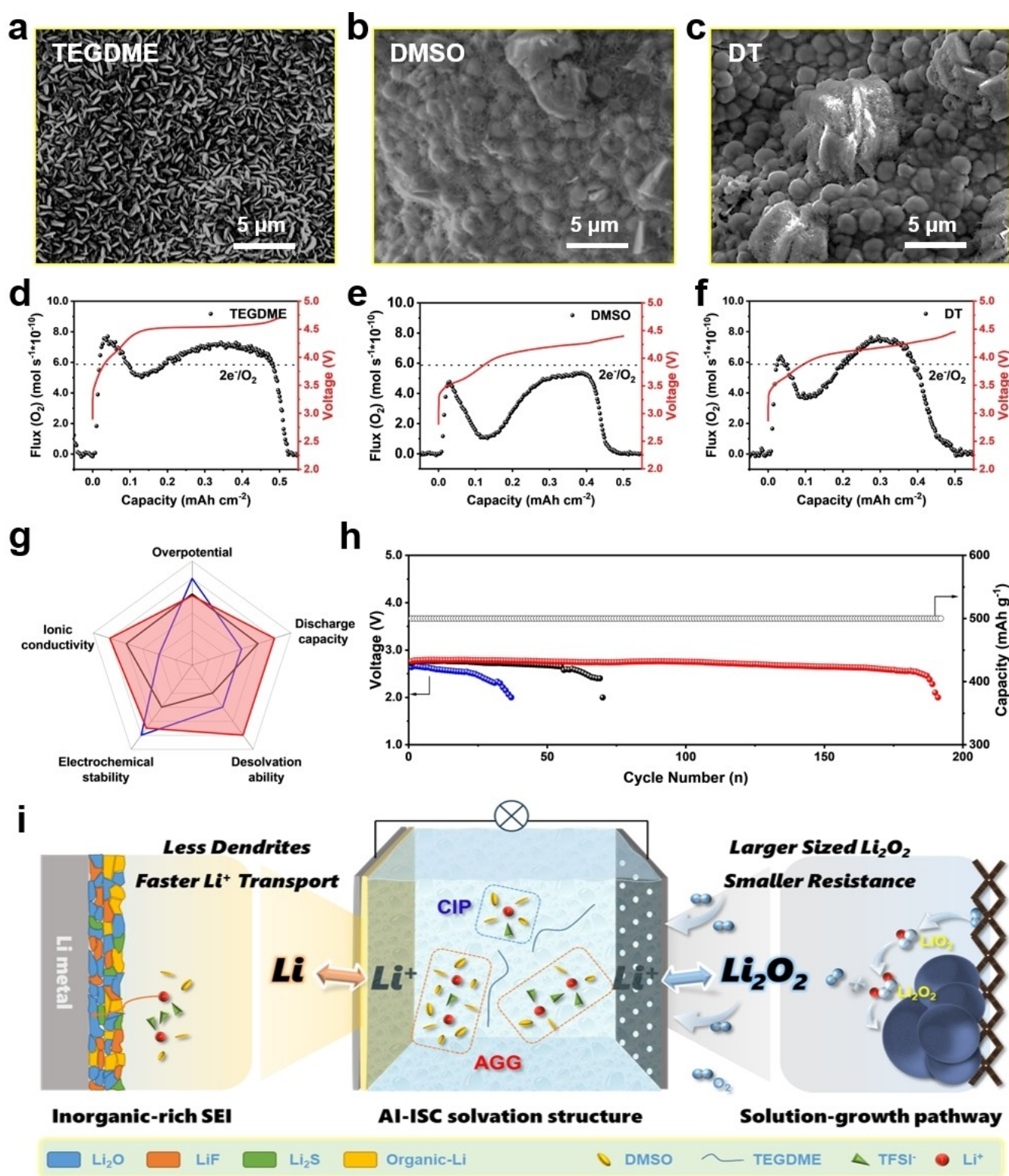


Figure 4. SEM images of cathodes after full discharge by using (a) TEGDME, (b) DMSO, and (c) DT electrolytes; In situ DEMS of LOBs by using (d) TEGDME, (e) DMSO, and (f) DT during charge; (g) Comparison of the properties of electrolytes; (h) Cycling performance of LOBs with a limited capacity of 500 mAh g^{-1} at 500 mA g^{-1} ; (i) Schematic working mechanism of the DT-based LOBs.

DMSO ones (Figures S21 and S22), which can stably work for over 190 cycles at a current density of 500 mA g^{-1} with a cutoff

capacity of 500 mAh g^{-1} , significantly outperforming 37 cycles of TEGDME and 70 cycles of DMSO (Figure 4h and S23–24).

In all, this study highlights the pivotal role of Li^+ solvation structures in optimizing LOB performance. Guided by machine learning, a dual-solvent electrolyte system (DMSO/TEGDME with LiTFSI) was successfully developed, achieving a delicate balance between the anode and cathode reactions. In the optimized electrolyte, it can show a promising AI-ISC solvation structure, which favors the formation of an inorganic-rich SEI, which greatly reduce interfacial impedance, improves Li^+ transport efficiency, and suppresses Li dendrite growth. Simultaneously, the fine modulation of Li^+ solvation/desolvation processes regulates the growth behavior of Li_2O_2 in favor of forming large sized Li_2O_2 particles, significantly improving the discharge capacity and cycling stability. These findings underscore the importance of precise electrolyte modulation, offering theoretical guidance for the development of advanced LOBs with high capacity, low overpotential, and prolonged cycle life.

Supporting Information

The authors have cited the additional reference within the Supporting Information.^[17]

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Bayesian optimization • electrolyte • $\text{Li}-\text{O}_2$ batteries • machine learning • SEI

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